

Heejun Shin

current as of July 2026

CONTACT	Department of Biostatistics Harvard T.H. Chan School of Public Health 655 Huntington Avenue Building 2 Room 435 Boston, MA 02115	<i>Email: heejunshin@hsph.harvard.edu</i> <i>Webpage: hshin111.github.io</i>
INTERESTS	Causal Inference, Bayesian Nonparametric Models, Environmental Statistics	
EMPLOYMENT	Postdoctoral Research Fellow , Harvard University <i>Mentor: Professor Francesca Dominici</i>	2024–present
	Statistical Consultant , Harvard Data Science Initiative “Real-Time AI Tool for Severe Event Prevention” in partnership with GE Vernova	2025–present
EDUCATION	Ph.D. in Statistics, University of Florida <i>Thesis: Causal Inference with Bayesian Modeling for Challenging Environmental Health Problems</i> <i>Advisor: Professor Joseph Antonelli</i>	2024
	B.S. in Applied Statistics, Konkuk University <i>Ranked first in class</i>	2019
PREPRINTS	Shin, H. , Braun, D., Irene, K., and Antonelli, J. (2023+). A spatial interference approach to account for mobility in air pollution studies with multivariate continuous treatments. arXiv:2305.14194. <i>Under Revision, Journal of the American Statistical Association</i> [Link] • <i>Winning paper of the 2025 ASA Epidemiology Section Norman Breslow Prize.</i> • <i>Winning paper of the 2025 ASA HPSS Section Student Paper Competition. (Declined)</i>	
PUBLICATIONS	[3] Antonelli, J., Shin, H. , Kang, S., Franks, A., Audirac, M., Braun, D. (2025) Robust Statistical Approaches to Understanding the Causal Effect of Air Pollution Mixtures. <i>Health Effects Institute</i> . [Link] [2] Shin, H. , Linero, A., Audirac, M., Irene, K., Braun, D., and Antonelli, J. (2025). Treatment effect heterogeneity and importance measures for multivariate continuous treatments. <i>The Annals of Applied Statistics</i> , 19(3): 1847-1867. [Journal][arXiv] • <i>Winning paper of the 2024 ASA Biometrics Section Early Career Paper Awards.</i> • <i>Winning paper of the 2024 ASA ENVR Section Student Paper Competition. (Declined)</i> [1] Shin, H. and Antonelli, J. (2023). Improved inference for doubly robust estimators of heterogeneous treatment effects. <i>Biometrics</i> , 79(4): 3140–3152. [Journal][arXiv] • <i>Winning paper of the 2022 ENAR John Van Ryzin Award.</i>	
HONORS & AWARDS	IMS New Researcher Travel Award , Institute of Mathematical Statistics 2025 The Norman Breslow Prize for the top paper among the winning papers of the Early Career Paper Awards, American Statistical Association Statistics in Epidemiology Section 2025 Early Career Paper Award , American Statistical Association Biometrics Section 2024 Kenneth and Janet Keene Endowed Dissertation Fellowship , College of Liberal Arts and Sciences, University of Florida 2024 Mark CK Yang Student Research Award , UF Joint Statistics and Biostatistics Workshop 2023 Boyd Harshburger Travel Award funded by National Science Foundation, The Southern Regional Council on Statistics 2022 The John Van Ryzin Award for the top paper among the winning papers of the Distinguished Student Paper Awards, International Biometric Society Eastern North American Region 2022 College of Liberal Arts and Sciences Graduate Student Travel Award , University of Florida - Joint Statistical Meetings 2023 - Eastern North American Region Spring Meeting, Joint Statistical Meetings 2022 The President’s Award for an outstanding undergraduate student, Korean Statistical Society 2019	

INVITED PRESENTATIONS	Seminar talk. “A Spatial Interference Approach to Account for Mobility in Air Pollution Studies with Multivariate Continuous Treatments” <i>London School of Hygiene & Tropical Medicine</i> . Virtual 2025.		
	Seminar talk. “Treatment effect heterogeneity and importance measures for multivariate continuous treatments” <i>Konkuk University</i> . Seoul, Korea 2024.		
	Seminar talk. “Improved inference for doubly robust estimators of heterogeneous treatment effects” <i>WebENAR</i> . Virtual 2024.		
	Seminar talk. “Identifying Interactions between Covariates and Continuous Exposures via Targeted Smoothing” <i>UFSTAT Student Seminar Series</i> . Gainesville, FL. 2023.		
CONTRIBUTED PRESENTATIONS	Contributed talk. “Data-driven Principal Stratification for Continuous Exposure and Mediators” <i>International Biometric Conference</i> . Seoul, Korea 2026.		
	Contributed poster. “Data-driven Principal Stratification for Continuous Exposure and Mediators” <i>The 2026 ISBA World Meeting</i> . Nagoya, Japan. 2026.		
	Contributed talk (as the Norman Breslow Prize winner). “A spatial interference approach to account for mobility in air pollution studies” <i>Joint Statistical Meetings</i> . Nashville, TN. 2025.		
	Contributed talk (as an Early Career Paper Award recipient). “Treatment effect heterogeneity and importance measures for multivariate continuous treatments” <i>Joint Statistical Meetings</i> . Portland, OR. 2024.		
	Contributed talk. “A spatial interference approach to account for mobility in air pollution studies with multivariate continuous treatments” <i>Bayesian Young Statisticians Meeting</i> . Virtual. 2023.		
	Contributed talk. “A spatial interference approach to account for mobility in air pollution studies with multivariate continuous treatments” <i>UF Joint Statistics and Biostatistics Workshop</i> . Gainesville, FL. 2023.		
	Contributed talk. “A spatial interference approach to account for mobility in air pollution studies with multivariate continuous treatments” <i>Joint Statistical Meetings</i> . Toronto, Canada. 2023.		
	Contributed poster. “Causal Effects of Continuous Exposures in the Presence of Spatial Interference: the Effects of Air Pollution on Public Health.” <i>American Causal Inference Conference</i> . Austin, TX. 2023.		
	Contributed poster. “Improved inference for doubly robust estimators of heterogeneous treatment effects.” <i>The Southern Regional Council on Statistics Summer Research Conference</i> . Jekyll Island, GA. 2022.		
	Contributed talk. “Improved inference for doubly robust estimators of heterogeneous treatment effects.” <i>Joint Statistical Meetings</i> . Washington, D.C. 2022.		
TEACHING EXPERIENCE	Guest Lecturer , Harvard University		2024
	BST228 - Applied Bayesian Analysis		
	Instructor , University of Florida		2023
	STA3024 - Introduction to Statistics II		
	Teaching Assistant , University of Florida		2019–2023
	STA3024 - Introduction to Statistics II		STA4322 - Introduction to Statistical Theory
	STA4241 - Statistical Learning		STA4502 - Nonparametric Statistical Methods
	STA4321 - Introduction to Probability		STA6166 - Statistical Methods in Research
MENTORING EXPERIENCE	Harvard University		
	Suhwan Bong, Ph.D. Student, Harvard University		2025–present
	Salome Kakhaia, visiting Ph.D. student, Utrecht University		2025–present
SOFTWARE	<i>SepBART</i> : An R package for estimating heterogeneous treatment effects of multivariate continuous exposures (https://github.com/hshin111/SepBART)		
SERVICE	Organizer, <i>UFSTAT Student Seminar Series</i> , Department of Statistics, University of Florida		2022-2023
	Paper Competition Judge, <i>ASA Section on Statistics in Epidemiology</i>		
	Journal Referee, <i>Annals of Applied Statistics</i> , <i>Biometrics</i> , <i>GeoHealth</i> , <i>Harvard Data Science Review</i> , <i>PeerJ: life and environment</i> , <i>Stat</i> , <i>Statistics in Biosciences</i> , <i>Statistics in Medicine</i>		

REFERENCES

Dr. Francesca Dominici

Clarence James Gamble Professor of Biostatistics,
Population, and Data Science
Harvard T.H. Chan School of Public Health
655 Huntington Avenue, SPH2, 4th Floor
Boston, Massachusetts, 02115
fdominic@hsph.harvard.edu • +1 (410) 258-5886

Dr. Joseph Antonelli

Associate Professor of Statistics
University of Florida
206 Griffin-Floyd Hall, P.O. Box 118545
Gainesville, FL 32611-8545
jantonelli@ufl.edu • +1 (352) 273-4638